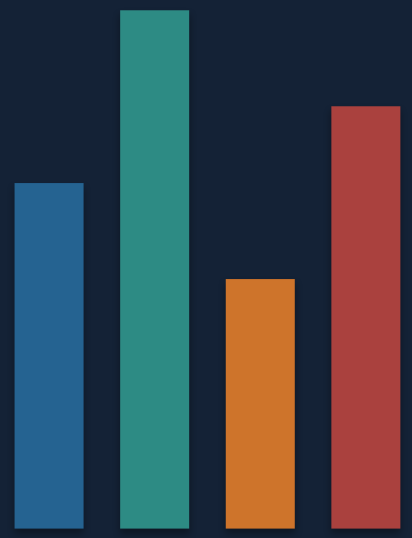
Long-Term Oil Price Forecast Evaluation

Are institutional Brent forecasts useful for upstream investment planning?



Berkeley Energy Project | Annual Brent Spot Price in real 2025 USD/bbl

Decision Context

Why forecast accuracy matters for upstream oil and gas firms

Upstream capital projects are large, long-lived, and partially irreversible. Oil price assumptions flow directly into DCF valuations.

Oil Price Forecast → **Revenue Projection** → **DCF / NPV / IRR** → **Investment Decision**

⚠ **Risk: Systematic overprediction encourages overinvestment and future asset impairments.**

- Large upfront capital commitments with multi-year payoff horizons
- Price assumptions drive revenue projections, NPV, and IRR
- Partially irreversible decisions — cost of over-approval is high

Research Questions

1

Does EIA beat a naive benchmark?

Does the institutional forecast outperform a simple random-walk carry-forward rule?

2

Is there systematic forecast bias?

Does EIA consistently overpredict or underpredict realized annual Brent prices?

3

Can simple models improve on EIA?

Do rolling AR models with or without futures data reduce forecast error further?

Horizons evaluated: 3-year and 5-year ahead forecasts of annual average Brent Spot Price in real 2025 USD/bbl.

Data and Scope

Building a reproducible forecast vintage panel

EIA AEO Table 12

Brent forecast vintages — primary evaluation target

Clean annual AEO releases from 2013–2023; no 2024 AEO was published.

FRED Brent (DCOILBRENTEU)

Realized Brent price — outcome variable

Daily spot prices aggregated to annual averages; CPI-deflated to real 2025 USD/bbl.

FRED CPI (CPIAUCSL)

Inflation deflator

Converts nominal EIA vintage forecasts to real 2025 USD/bbl for consistent comparison.

Yahoo Finance BZ=F

Front-month Brent futures — market signal

Used as an additional regressor in the rolling AR + Futures extension.

Note: World Bank Crude oil (avg) used as supplementary check only — different contract from Brent spot.

Evaluation Pipeline

Standardizing forecasts and realized prices into one panel

EIA Vintages
AEO Table 12 forecast releases

→

Realized Brent
FRED daily spot → annual average

→

CPI Deflation
Nominal → Real 2025 USD/bbl

→

Panel
3y & 5y horizon outcome pairs

→

Benchmark
Random-walk comparison

EIA Publication Vintages

11

evaluation windows

Forecast-Outcome Pairs

20

3-year and 5-year

Price Unit

2025 $

real USD / bbl

Evaluation Design

Metrics and benchmark definition

Forecast Error Definition

Forecast error = forecast − realized

- ME (Mean Error) — average bias; positive = optimistic
- MAE — mean absolute magnitude of error
- RMSE — root mean squared error
- MAPE — error as share of realized price

Random-Walk Benchmark

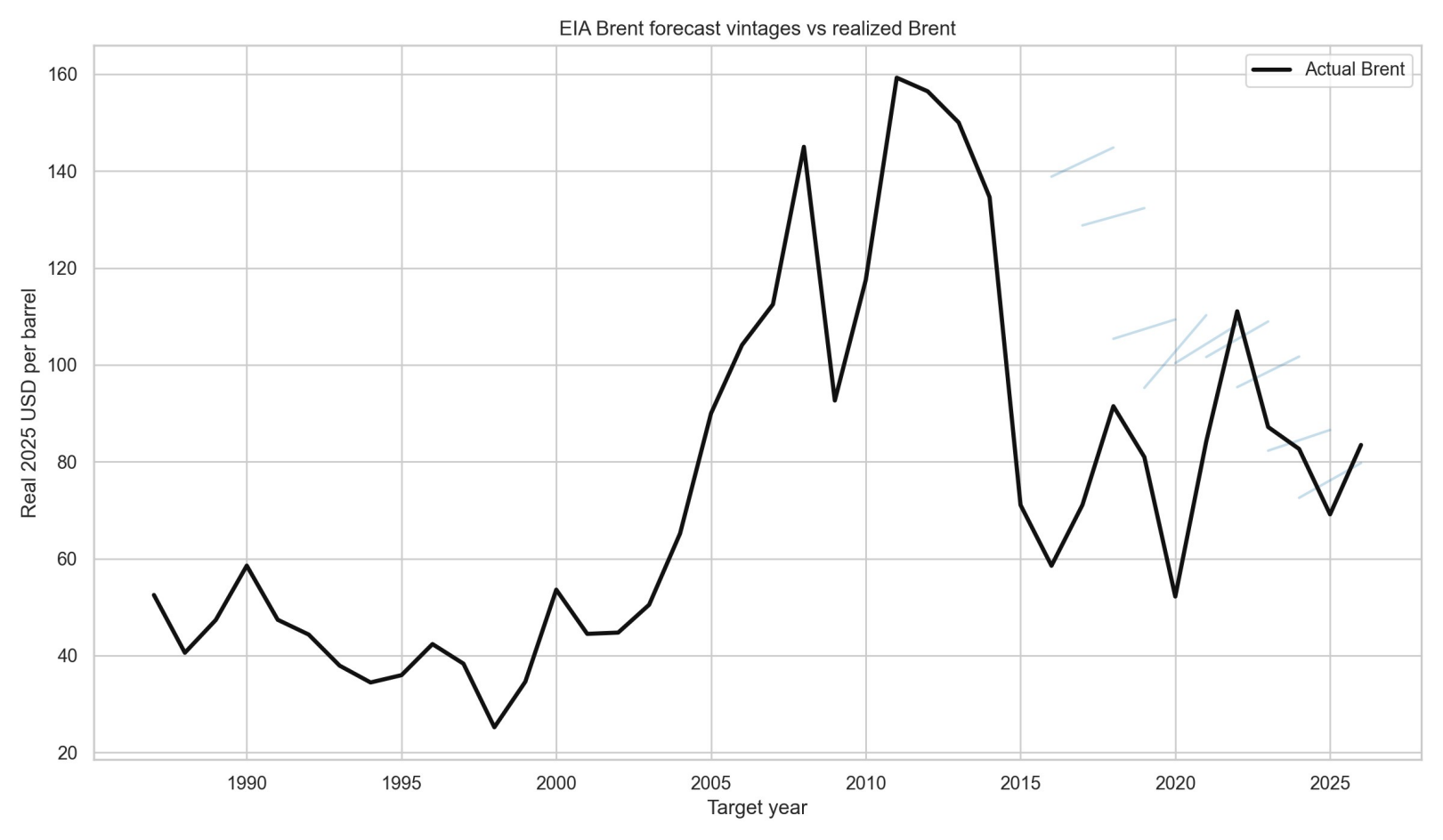
Forecast = most recently completed annual Brent average available at time of release.

Simple, transparent, and the standard commodity-price baseline.
If EIA beats this, it demonstrates genuine informational value.

Simple · Transparent · Hard to beat

EIA Forecast Vintages vs Realized Brent

Institutional forecasts follow smooth paths; realized prices are far more volatile



Takeaway: EIA vintages are smooth, mean-reverting trend lines; realized Brent is highly volatile — motivating formal accuracy and bias evaluation.

EIA Beats the Random Walk

EIA forecasts contain real information relative to a carry-forward rule

3-Year RMSE

EIA		Random Walk
34.97	vs	**42.99**

EIA is 8.02 lower (19% reduction)

5-Year RMSE

EIA		Random Walk
34.32	vs	**47.34**

EIA is 13.02 lower (28% reduction)

- EIA also beats random walk on MAE: 26.01 vs 31.66 (3-year); 28.09 vs 38.90 (5-year).
- Finding: EIA long-term Brent forecasts are informative — they outperform a simple carry-forward rule.

But EIA Is Optimistically Biased

Positive mean error indicates systematic overprediction of realized Brent

3-Year Mean Error

+20.44

USD/bbl average overprediction

5-Year Mean Error

+26.64

USD/bbl average overprediction

- Positive ME means EIA forecasts are above realized Brent on average — optimistically biased.
- Bias compounds in DCF: inflated price → inflated revenue → inflated NPV and IRR.
- Implication: EIA is informative, but should not be used mechanically without bias adjustment.

Capital Budgeting Risk

Higher forecast prices inflate projected revenue, NPV, and IRR — creating a systematic bias toward over-approval of upstream projects.

Risk: Overinvestment

Rolling Model Extension

Simple econometric extensions do not improve on EIA

Model	3-Year RMSE	5-Year RMSE
EIA	34.97	34.32
Rolling AR	48.95	51.24
Rolling AR + Futures	52.51	53.23

Model Details

- Rolling AR: autoregression on lagged annual Brent
- AR + Futures: adds front-month BZ=F futures and basis spread
- Estimated on expanding window; predicted out-of-sample

Finding: Neither rolling extension beats EIA. EIA remains the best available forecast in this comparison.

Interpretation

Useful, but not unbiased

Informative relative to naive benchmark

- EIA beats random walk on RMSE and MAE at both horizons.
- Institutional forecasts add real informational content.
- Dismissing EIA forecasts entirely would be premature.

Biased enough to matter for investment

- Consistent positive ME of $20–$27/bbl at 3- and 5-year horizons.
- Bias flows directly into revenue and valuation models.
- Mechanical use of EIA price decks risks overinvestment.

Synthesis: EIA is informative but biased — use it as a starting point, not a final answer.

Managerial Implications

How upstream firms should use institutional price decks

01

Use EIA as a baseline, not a final answer

EIA beats the random walk — discarding it entirely would discard genuine information.

02

Apply a bias adjustment or conservative haircut

Historical ME: –$20/bbl at 3-year; –$27/bbl at 5-year. Run sensitivities around this.

03

Run downside price scenarios before final investment decisions

Stress-test projects against plausible realized price paths, not smooth forecast paths.

04

Treat long-horizon price uncertainty as a capital allocation risk

Forecast uncertainty should appear in project risk registers and inform hurdle-rate calibration.

Limitations and Next Steps

Scope boundaries and natural extensions

Limitations

- Main analysis covers EIA; IEA and other institutions not compared.
- World Bank uses Crude oil (avg), not Brent spot — different contract.
- Futures extension uses front-month BZ=F, not a long-dated forward curve.
- Long-horizon realized sample is naturally limited (~10 –11 outcomes).

Next Steps

- Add IEA or World Bank as multi-institution robustness check.
- Test bias-corrected EIA: apply historical ME adjustment to future vintages.
- Use long-dated Brent futures curve if reliable historical data are available.
- Extend panel as new EIA AEO vintages are published annually.

Conclusion

EIA long-term Brent forecasts are informative, but optimistically biased; upstream firms should adjust them before using them in capital budgeting.

Informative

EIA beats random walk — use it as a baseline

Biased

ME of +$20–$27/bbl — apply a haircut

Actionable

Adjust before DCF — run downside scenarios

Use institutional forecasts as starting points, not unadjusted truth.

Appendix A: Full Benchmark Metrics

Model	Horizon	N	ME	MAE	RMSE	MAPE
EIA	3	11	20.44	26.01	34.97	0.388
EIA	5	9	26.64	28.09	34.32	0.381
Random Walk	3	11	17.25	31.66	42.99	0.438
Random Walk	5	9	13.79	38.90	47.34	0.512

Appendix B: Rolling Model Metrics

Model	Horizon	N	ME	MAE	RMSE	MAPE
Rolling AR	3	26	10.13	35.33	48.95	0.447
Rolling AR	5	24	-5.41	35.95	51.24	0.379
Rolling AR + Futures	3	9	17.96	40.11	52.51	0.549
Rolling AR + Futures	5	7	7.57	36.61	53.23	0.539

Appendix C: Data Source Links

- EIA Annual Energy Outlook Table 12 — eia.gov/aeo
- FRED DCOILBRENTEU: Brent Spot Price — fred.stlouisfed.org
- FRED CPIAUCSL: Consumer Price Index — fred.stlouisfed.org
- Yahoo Finance BZ=F: Brent crude front-month futures proxy
- World Bank Commodity Markets price archive — worldbank.org

Appendix D: Reproducibility

- Structured Python package: src/oil_forecast_project
- Pipeline writes CSVs, charts, Excel workbook, and summary automatically
- Tests cover metrics, CPI conversion, Brent aggregation, panel construction, and benchmark summary
- Validation: pytest passes with 6 tests